DAY 1

Session 1 (10:00 – 11:30)

Business Intelligence & Power BI Basics

Topics:

- What is Business Intelligence
- Role of Power BI in decision making
- Power BI ecosystem (Desktop, Service, Mobile)
- Interface overview (Report / Data / Model)
- Importing data from Excel

Participants will:

- Import dataset
- Explore tables
- Create first visual (Sales by Category)

Session 2 (11:45 – 1:15)

Data Preparation with Power Query

Topics:

- Power Query Editor
- Data cleaning
- Removing blanks & duplicates
- Changing data types
- Splitting columns
- Renaming columns

Tasks:

- Remove duplicates
- Split full name column
- Convert salary to currency
- Clean department names

Session 3 (2:00 – 3:30)

Data Modeling

Topics:

- Fact vs Dimension tables
- Relationships
- Cardinality
- Star schema

Tables:

- Orders
- Products
- Customers
- Date

Participants will:

- Create relationships
- Build a star schema
- Understand filter direction

Session 4 (3:45 – 5:15)

DAX Fundamentals

Topics:

- Introduction to DAX
- Calculated columns vs Measures
- Basic functions:
    - SUM
    - AVERAGE
    - COUNT
    - DISTINCTCOUNT
    - FILTER
    - ALL
    - ALLSELECTED

Practical Exercise
Participants create:

Measures:

- Total Sales
- Total Profit
- Profit Margin %

DAY 2

Session 5 (10:00 – 11:30)

Time Intelligence

Topics:

- Date table
- Time intelligence concepts

Functions:

- TOTALYTD
- SAMEPERIODLASTYEAR

Practical Exercise

Participants build:

- Sales YTD
- Sales Last Year
- Growth %

Visual
Line chart for monthly sales.

Session 6 (11:45 – 1:15)

Power BI Visualizations

Topics:

- Tables
- Matrix
- Cards
- KPI visuals
- Bar / Column / Line / Pie charts

Practical Exercise
Participants build a **Sales Dashboard**

Visuals:

- Sales by Region
- Sales by Category
- Top Products
- Profit Margin KPI

Session 7 (2:00 – 3:30)

Interactivity & Dashboard Design

Topics:

- Slicers
- Page filters
- Report filters
- Drill-down
- Drill-through
- Visualization best practices
- Business storytelling

Practical Exercise
Participants build **interactive dashboards**

Features:

- Region slicer
- Year filter
- Drill-down from Category → Product

Session 8 (3:45 – 5:15)

Power BI Service & Advanced Integration

Topics:

- Publish to Power BI Service
- Scheduling refresh
- Query folding
- DAX optimization
- Mobile layout
- JSON & REST API
- Connecting to databases
    - MySQL
    - PostgreSQL